An Unconditional Proof of the Unboundedness Conjecture in OEIS A358200

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Abstract

The Online Encyclopedia of Integer Sequences (OEIS) entry A358200 records the conjecture that a frequency-ranking sequence associated with ratios of consecutive prime gaps is unbounded. We show unconditionally that the stronger identity $a(n) = n$ holds for infinitely many indices. In fact, there are infinitely many first occurrences of ratios that establish new record highs, and infinitely many establishing new record lows. The proof is an immediate application of János Pintz's established theorem that successive prime-gap ratios have limit superior $+\infty$ and limit inferior 0. The note isolates this consequence for OEIS A358200; it does not claim a new theorem about prime gaps.

1 Definitions and claim

Write $p_1 = 2 < p_2 = 3 < \cdots$ for the primes, and define the successive gaps

$$d_n = p_{n+1} - p_n \qquad (n \geq 1).$$

For $n \geq 3$, set

$$r(n) = \frac{p_n - p_{n-1}}{p_{n-1} - p_{n-2}} = \frac{d_{n-1}}{d_{n-2}}.$$

The OEIS sequence A358200, denoted here by $a(n)$ for $n \geq 4$, assigns the frequency-rank of $r(n)$ among the ratios $r(3), \ldots, r(n-1)$ if that ratio has already appeared. The previous ratios are ordered by decreasing frequency and, in the case of a tie, by increasing numerical value. Crucially, when $r(n)$ has *not* appeared previously, the OEIS definition sets

$$a(n) = n. \tag{1}$$

This convention, rather than the precise tie-breaking rule, is all that is needed below [1].

Theorem 1 (Resolution of the unboundedness assertion in A358200)**.** *For the sequence $a(n)$ in OEIS A358200,*

$$\#\{n \geq 4 : a(n) = n\} = \infty.$$

Consequently,

$$\boxed{\limsup_{n\to\infty} a(n) = +\infty.}$$

The occurrences $a(n) = n$ include infinitely many indices where $r(n)$ is a new record maximum and infinitely many where it is a new record minimum.

2 The input from prime-gap theory

The following unconditional result was proved by Pintz [2, 3]:

$$\limsup_{j\to\infty} \frac{d_{j+1}}{d_j} = +\infty, \qquad \liminf_{j\to\infty} \frac{d_{j+1}}{d_j} = 0. \tag{2}$$

Both statements concern *consecutive* prime gaps; no unproved hypotheses, such as RH or GRH, are assumed. Equation (2) implies the corresponding assertions for the index-shifted sequence $r(n)$:

$$\limsup_{n\to\infty} r(n) = +\infty, \qquad \liminf_{n\to\infty} r(n) = 0. \tag{3}$$

3 Proof

Proof of Theorem 1. First consider record maxima. Suppose, for contradiction, that there are only finitely many indices $n \geq 4$ such that

$$r(n) > \max_{3\leq i<n} r(i).$$

Then the sequence of record maxima would eventually stop increasing; in particular $r(n)$ would be bounded above by the largest ratio observed before the final record. This contradicts $\limsup r(n) = +\infty$ in (3). Hence there are infinitely many new record-high ratios.

Similarly, suppose that only finitely many indices $n \geq 4$ satisfy

$$r(n) < \min_{3\leq i<n} r(i).$$

Because every prime gap is positive, the minimum of the finite set of ratios up to the final record is a positive number. After that final record, every ratio would be at least this positive minimum, contradicting $\liminf r(n) = 0$ in (3). Therefore new record-low ratios also occur infinitely often.

Every new record high or low is necessarily a ratio not previously encountered. By the exact OEIS rule (1), at each such index n we have $a(n) = n$. Thus

$$\#\{n \geq 4 : a(n) = n\} = \infty.$$

Along this infinite set of indices we have $a(n) = n \to \infty$, so $\limsup_{n\to\infty} a(n) = +\infty$. This proves all assertions. □

4 Equivalent first-occurrence formulation

Corollary 2. *The set of ratios of consecutive prime gaps is infinite:*

$$\#\left\{\frac{p_n - p_{n-1}}{p_{n-1} - p_{n-2}} : n \geq 3\right\} = \infty.$$

There are infinitely many distinct such ratios greater than 1 *and infinitely many distinct such ratios smaller than* 1*.*

Proof. The record-high ratios form an infinite strictly increasing sequence, and by (3) its values eventually exceed 1. Likewise, the record-low ratios form an infinite strictly decreasing positive sequence with values eventually below 1. □

5 Scope and provenance

The resolved assertion is the *unboundedness conjecture* listed as item 2 in the comments for OEIS A358200 [1]. The other conjectural statements in that entry—concerning a numerical ranking bound and limiting relative frequencies—are *not* proved here.

The main input is Pintz's theorem from 2014 (subsequently published in 2015), predating the creation of OEIS A358200 in 2023. Accordingly, this note presents a transparent, unconditional *deduction from a known theorem*, not a novel prime-gap estimate or a claim of priority. It may be useful as a documented citation for correcting the conjecture annotation in OEIS.

References

[1] A. Cicuttin, *OEIS A358200: Frequency ranking position of the ratio between consecutive prime gaps*, The On-Line Encyclopedia of Integer Sequences, entry created 22 February 2023. https://oeis.org/A358200.

[2] J. Pintz, *On the ratio of consecutive gaps between primes*, arXiv:1406.2658, 2014. https://arxiv.org/abs/1406.2658.

[3] J. Pintz, *On the ratio of consecutive gaps between primes*, in *Analytic Number Theory*, ed. C. Pomerance and M. Th. Rassias, Springer (2015), pp. 285–304. doi:10.1007/978-3-319-22240-0_17.

AI Authorship and Verification Disclosure

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